

Operating Systems — Reference Handbook

Operating Systems (OS) · Unit 1 · Introduction & Process Management · 2024

Overview

These notes cover **Operating Systems**. How operating systems manage processes, CPU time, memory, storage and devices — and keep it all safe and concurrent. Each section states the core idea, a worked intuition and the points examiners look for.

1. Operating System Structures

Monolithic, layered, microkernel and modular designs trade performance for isolation.

Why it matters: Operating System Structures appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of operating system structures in one line.
- A small example showing operating system structures in practice.
- Common mistakes students make and how to avoid them.

2. System Calls

System calls are the interface through which user programs request kernel services.

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- Key definition of system calls in one line.
- A small example showing system calls in practice.
- Common mistakes students make and how to avoid them.

3. Process Concept

A process is a program in execution with its own PCB, address space and state.

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- Key definition of process concept in one line.
- A small example showing process concept in practice.
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4. Scheduling Criteria

CPU utilisation, throughput, turnaround time, waiting time and response time evaluate schedulers.

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5. FCFS, SJF and Round Robin

FCFS is simple but suffers convoy effects; SJF minimises average waiting time; round robin uses time quanta.

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- Key definition of fcfs, sjf and round robin in one line.
- A small example showing fcfs, sjf and round robin in practice.
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6. Priority Scheduling

Priority scheduling can starve low-priority processes, which aging solves.

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- Key definition of priority scheduling in one line.
- A small example showing priority scheduling in practice.
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7. Deadlock Characterization

Deadlock requires mutual exclusion, hold and wait, no preemption and circular wait simultaneously.

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- Key definition of deadlock characterization in one line.
- A small example showing deadlock characterization in practice.
- Common mistakes students make and how to avoid them.

8. Resource Allocation Graph

A cycle in a single-instance resource allocation graph implies deadlock.

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- Key definition of resource allocation graph in one line.
- A small example showing resource allocation graph in practice.
- Common mistakes students make and how to avoid them.

9. Deadlock Prevention

Prevention denies at least one of the four necessary conditions, e.g. by ordering resources.

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- Key definition of deadlock prevention in one line.
- A small example showing deadlock prevention in practice.
- Common mistakes students make and how to avoid them.

10. Contiguous Allocation

First fit, best fit and worst fit allocate memory partitions, causing external fragmentation.

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- Key definition of contiguous allocation in one line.
- A small example showing contiguous allocation in practice.
- Common mistakes students make and how to avoid them.

11. Paging

Paging divides memory into fixed-size frames and uses page tables to translate addresses.

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- Key definition of paging in one line.
- A small example showing paging in practice.
- Common mistakes students make and how to avoid them.

12. Translation Lookaside Buffer

The TLB caches page table entries to make address translation fast.

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- Key definition of translation lookaside buffer in one line.
- A small example showing translation lookaside buffer in practice.
- Common mistakes students make and how to avoid them.

13. File Concepts

Files have attributes, operations and access methods such as sequential and direct access.

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- Key definition of file concepts in one line.
- A small example showing file concepts in practice.
- Common mistakes students make and how to avoid them.

14. Directory Structures

Single-level, two-level, tree and acyclic-graph directories organise files.

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- Key definition of directory structures in one line.
- A small example showing directory structures in practice.
- Common mistakes students make and how to avoid them.

15. Allocation Methods

Contiguous, linked and indexed allocation trade fragmentation against access speed.

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- A small example showing allocation methods in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Operating System Structures · System Calls · Process Concept · Scheduling Criteria · FCFS, SJF and Round Robin · Priority Scheduling · Deadlock Characterization · Resource Allocation Graph · Deadlock Prevention · Contiguous Allocation · Paging · Translation Lookaside Buffer · File Concepts · Directory Structures · Allocation Methods